AI OPEN
DATA CENTER



Automation-Native Infrastructure Design: from Light to Compute



Dave Evans *Mechatronics Engineer, Google*

Mike Mckee-Hector *Hardware Engineer, Microsoft*

Ryan Olson *Product Design Engineer, Meta*

Eric Xu *Sr. Robotics ENG Manager, Meta*

Panel Discussion



Dave Evans

Mechatronics Engineer, Google



Mike McKee-Hector

Hardware Engineer, Microsoft



Ryan Olson

Product Design Engineer, Robotics, Meta Platforms Inc.

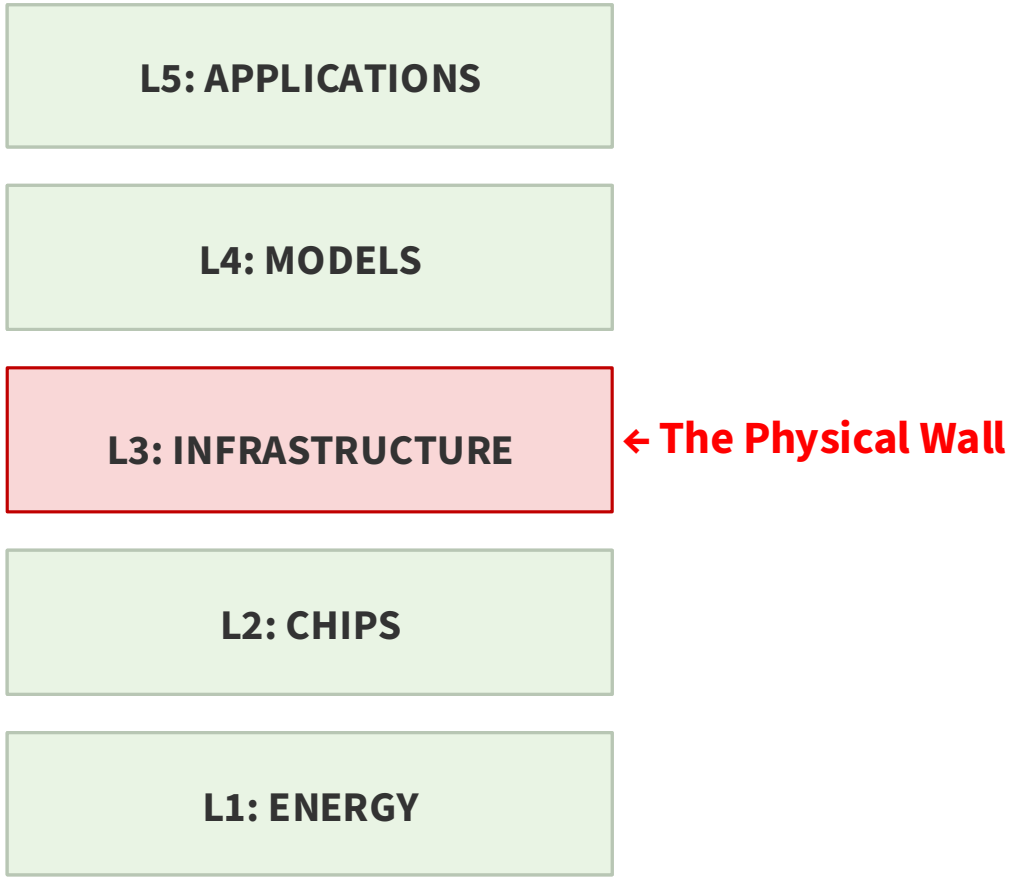


Eric Xu

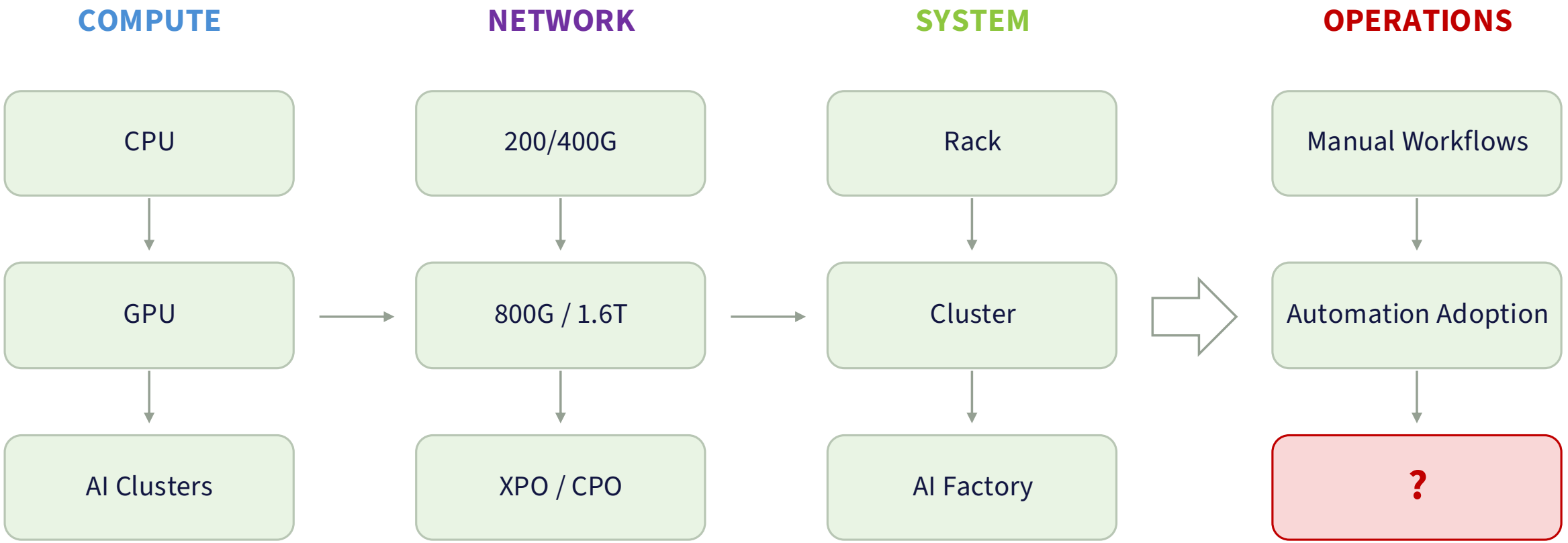
Sr. Robotics ENG Manager, Meta Platforms Inc.

The AI Stack Has a Physical Bottleneck

- In data centers, the flow of **bits** (compute, data, networking) is industrialized to near light speed.
- But the flow of **atoms** (physical operations) remains fragmented and manual.
- Rapid evolution of bits is pushing atoms beyond human scale — our hardware operations are hitting the physical wall.



The Next Frontier: Automation-Native Operations



Automation Guiding Principles

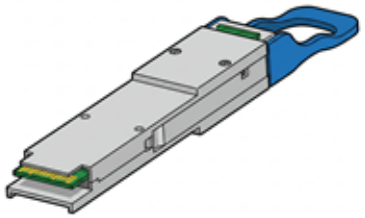
Design for automation is common, however the data center is a new domain

- Best design practices to design for automation
  - **Increase access** to visual reference features, labeling and touch points
  - **Reduce and simplify** process steps, range of motion, and degrees of freedom
  - Design for task based localization
    - Design for 3mm accuracy (general tolerance)
  - **Rigid touch points** wherever possible
  - **Blind mate** connections

Optics Roadmap - Industry Trends

Persistent Standards

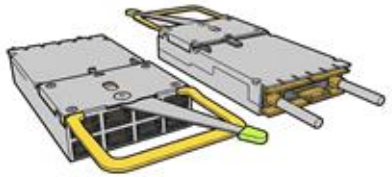
Legacy → USB-A, USB-C, RJ45
Fiber → LC,MPO



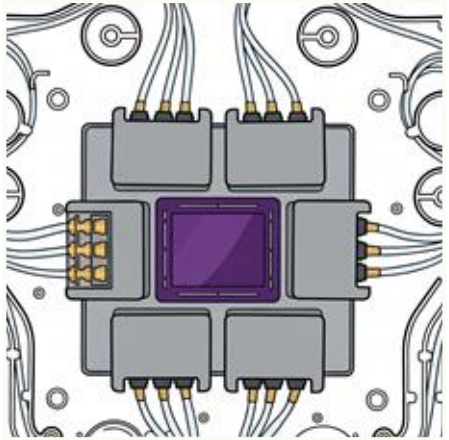
QSFP



OSFP



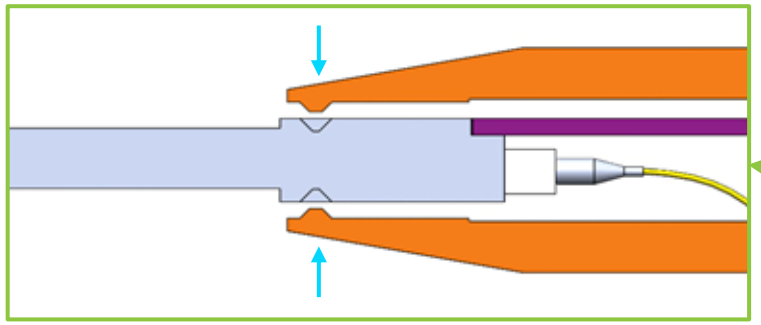
XPO

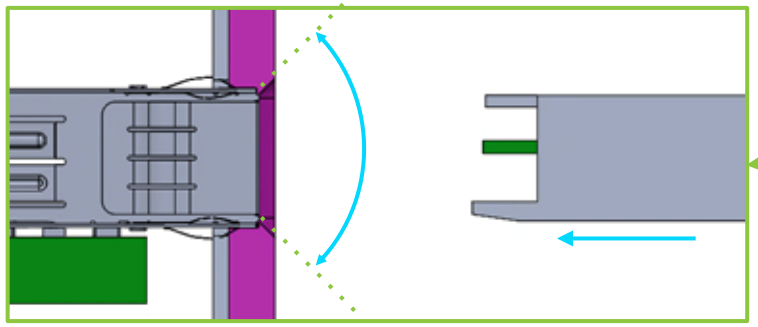


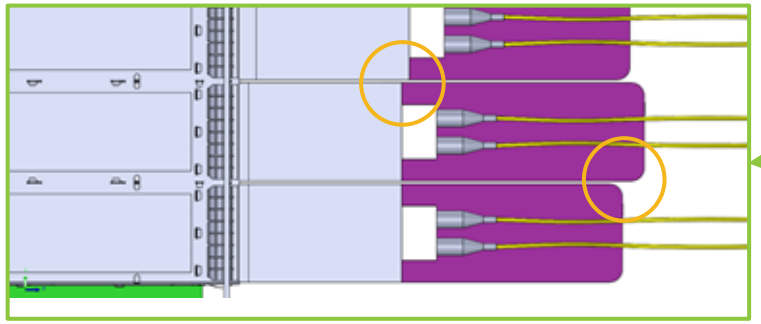
NPO/CPO

We are here

Optics Designs Recommendations





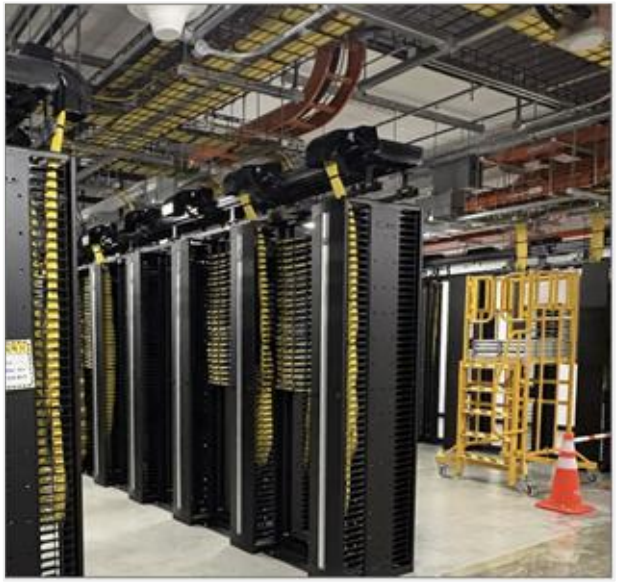


Call to industry for automation standardization

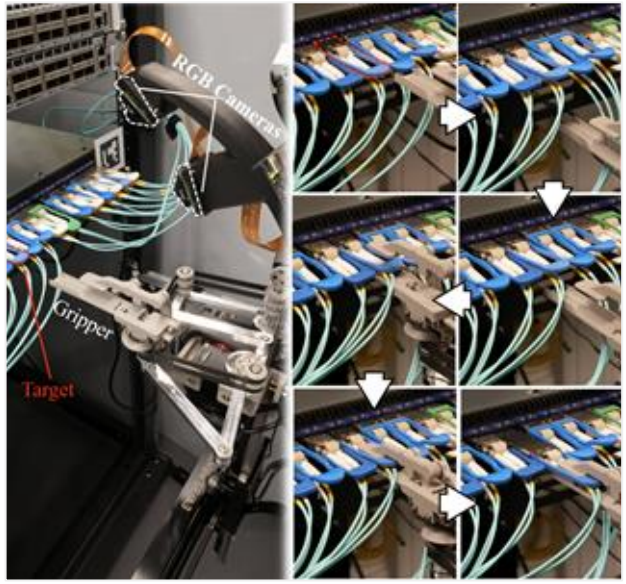
- **Robust grip locations**
- **Labeling standards**
- **Lead-ins**
- **Standardized and rigid pull tabs**
- **Positive seating verification**
- **Optimized datums**
- **Cable management for consistency and predictability**

Automation-Native Design in AI Deployment

Construction (backbone, pathway)	Provisioning (optics, fiber patching)	Deployment Validation (inspection, test)	Service & Repair (inspect, clean, fix)	Optics RMA (inspect, clean)



Construction (image from Meta)



Automated Optics Provisioning (image from Microsoft)



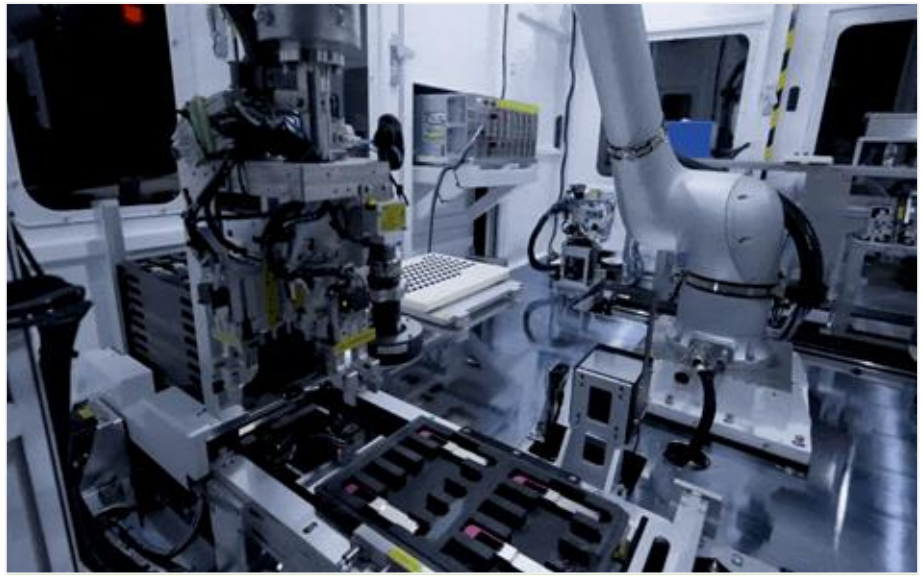
Fiber Provisioning (image from Meta)

Automation-Native Design in Service & Repair

Construction (backbone, pathway)	Provisioning (optics, fiber patching)	Deployment Validation (inspection, test)	Service & Repair (inspect, clean, fix)	Optics RMA (inspect, clean)



Field Service & Repairs (image from Meta)



Automated Optics RMA (image from Meta)

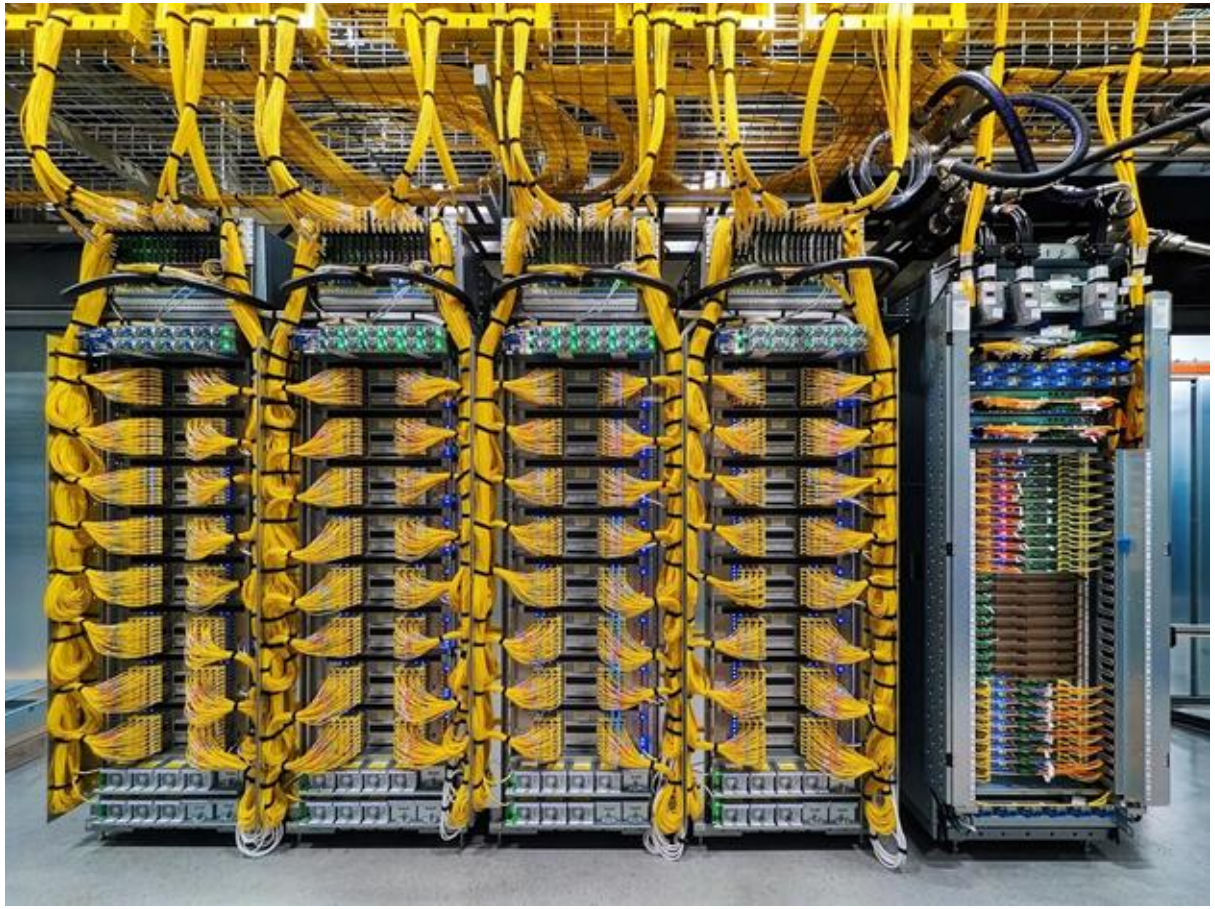
Trends and Future of Tray Design

- General trends in server design
  - **Compute density**
  - Liquid cooled trays require **quick-disconnect fittings** and **handles** for fluid connections
  - **Larger, heavier designs** lead to ergonomic risks
- Future trends
  - **Wider trays**
  - **FRUs** (Field Replaceable Units)



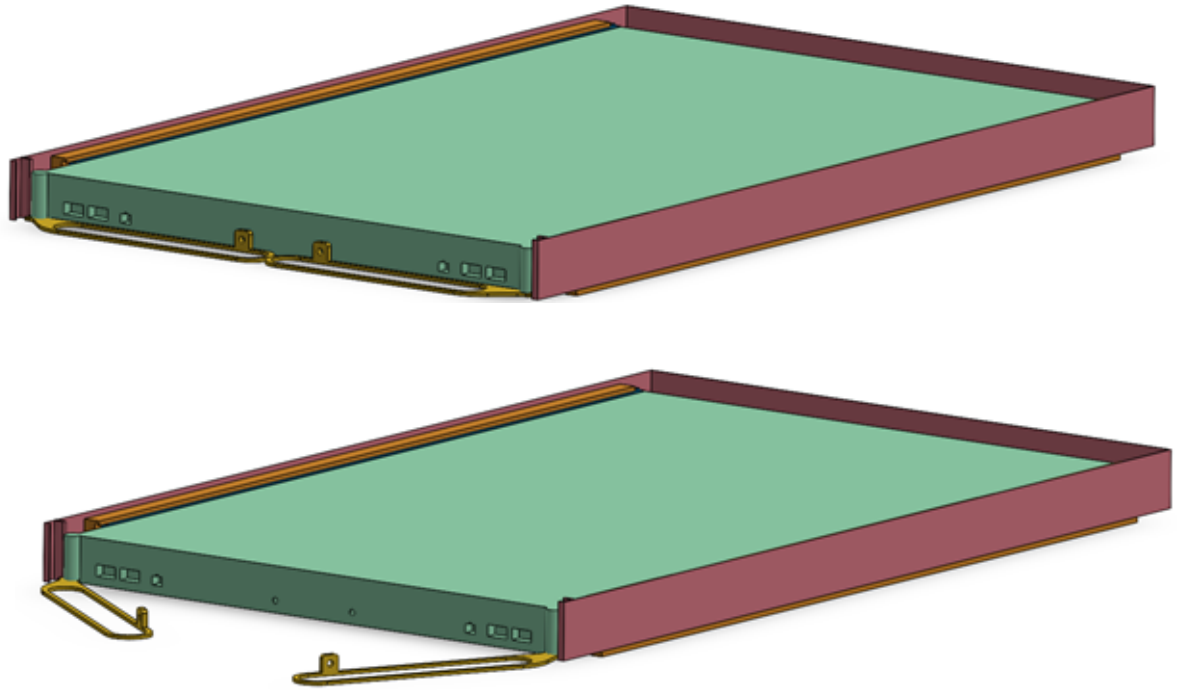
Tray Design Challenges

- **Cable Management**
  - Hide references, block touch points
- **Handles**
  - Shape variety
- **Retention**
  - Handle Swing
- **Rack-Tray Interfaces**
  - Installation tolerance
  - Retention



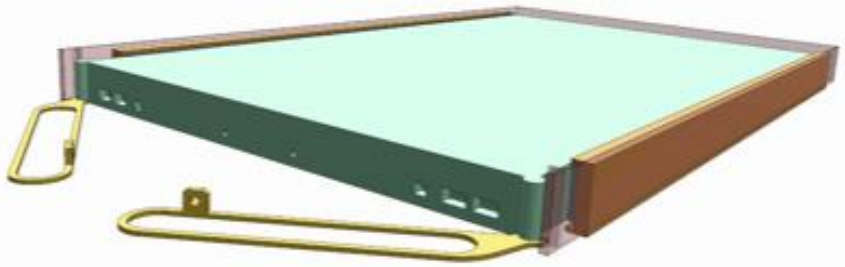
Tray Handles for Robot Interaction

- **Coordinated motion** is hard
- Minimize **degrees of freedom**
  - Handle engagement
  - Handle retention engage/release
- Minimize **handle swing angle**
- **Visual references** help robots align
- Add ways to **verify installation** is correct

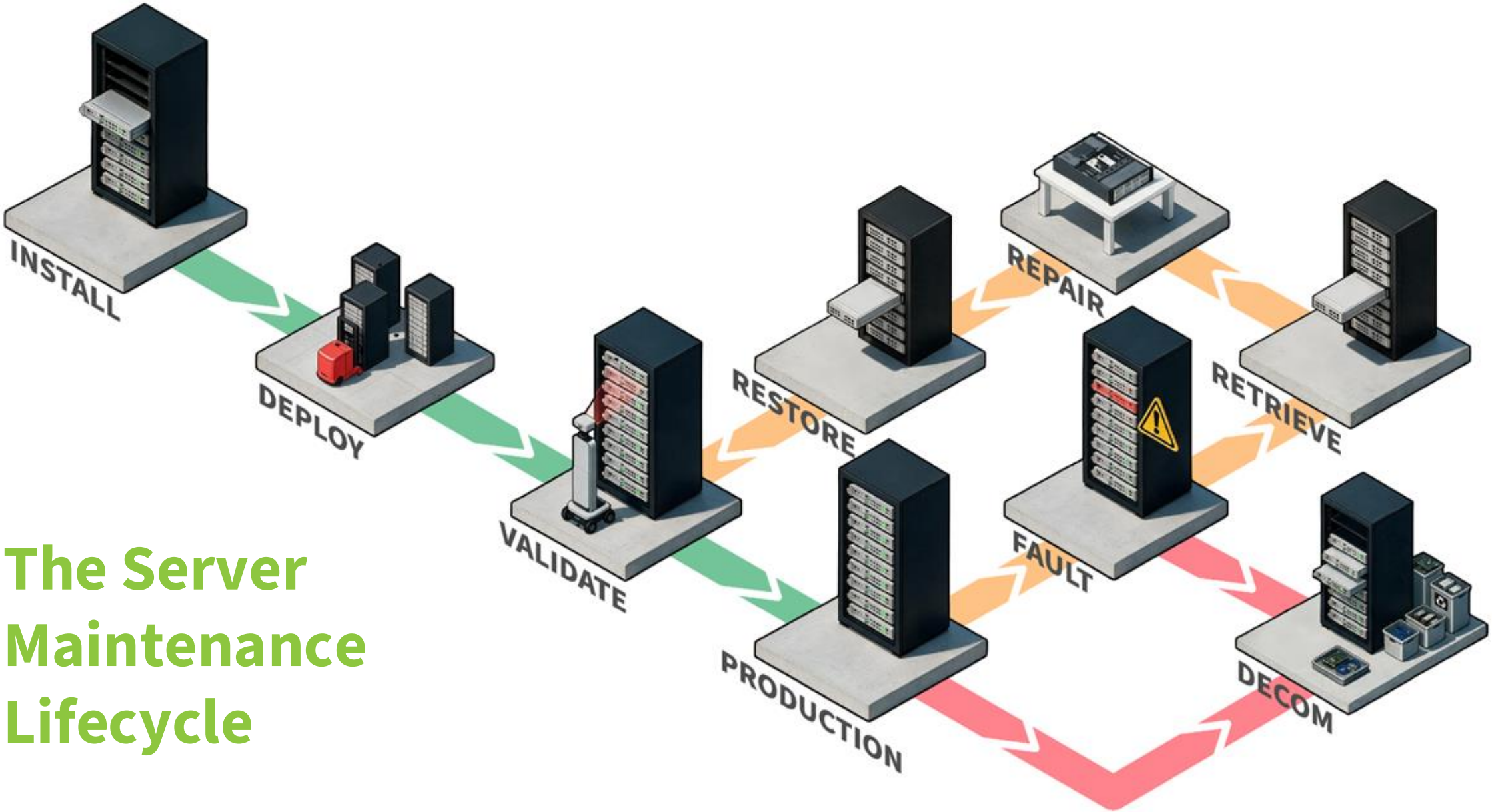


Rack-Tray Interfaces

- **Installation**
  - Locating feature for rack position
  - Lead-ins
  - Passive retention features
- **Removal**
  - Reduce degrees of freedom for retention features
  - Keep retention feature near front of tray

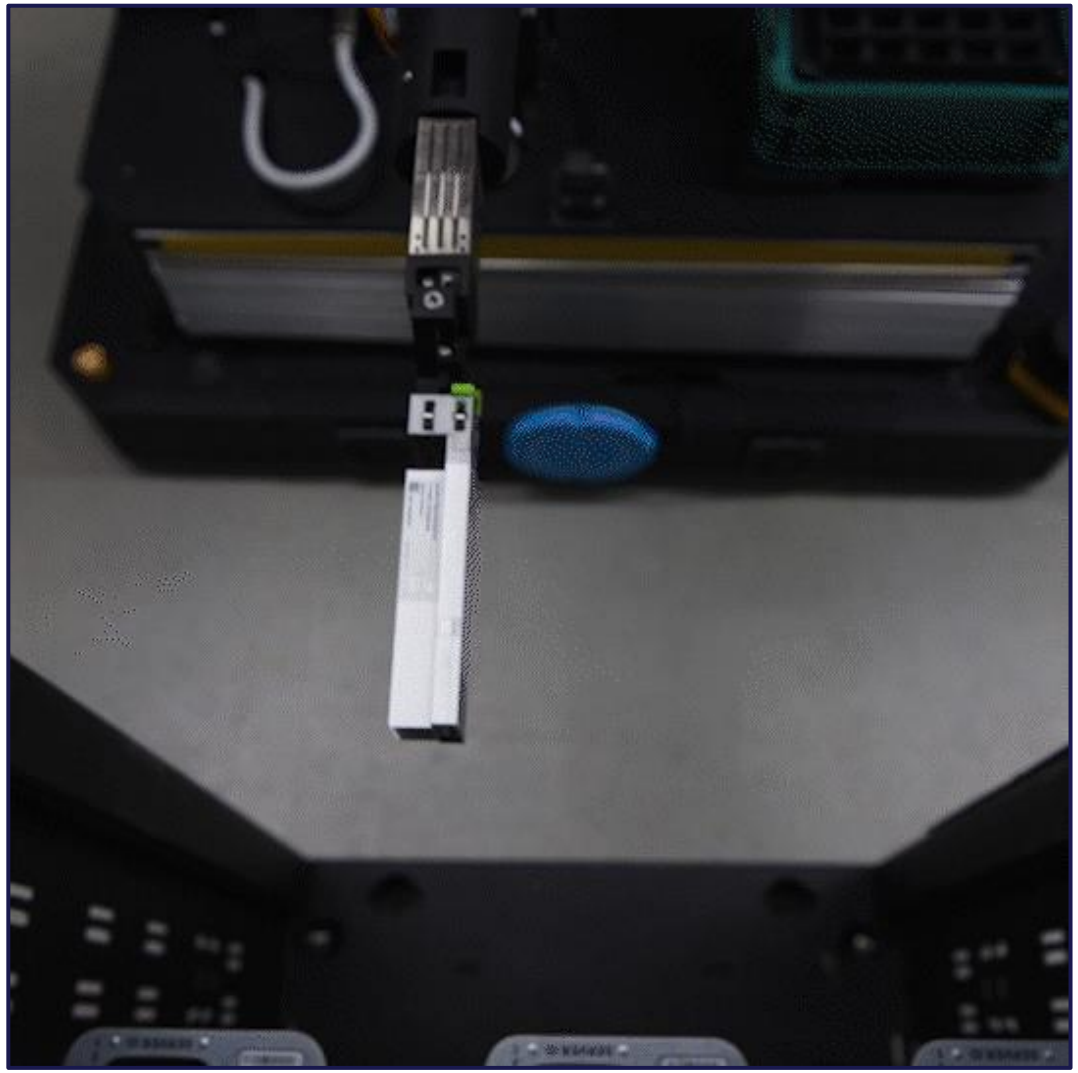


The Server Maintenance Lifecycle



Enabling Automated Assembly/Disassembly

- **Interconnect and tray design** improvements **enable automation** in operations
- **Complexity** of robotic solutions depends on **design decisions** made upstream



Meta

Automated Inspection and Validation

- Machine **inspection** and **validation** at multiple points of operation
  - Post-deployment **commissioning and QA**
  - Ongoing **health monitoring**
  - Fault **verification and response**
  - **Post repair** validation



Meta

Bridging Human and Automated Operations

- Automation-ready design must support both **human operators and robots**
  - There will always be people in the loop
  - Standards should help both people and automation
  - Must consider safety and ergonomics
- Robotics as a **foundational infrastructure layer** alongside power, cooling, connectivity, and compute with shared interfaces and dual operability



Meta

Call to Action

We work with leading robotics and automation partners to advance safer, standardized DC automation solutions.

- We help scale proven, real-world solutions — and welcome teams to connect with us at automation@ocproject.net
- Stay tuned for two automation-native design white papers
  - Optics & Fiber Automation-Native Design
  - Server Automation-Native Design

automation@ocproject.net

Thank You!

